Georeferencing: Castra Vetera

geo0240 · a photographed map and a reconstruction drawing in today's coordinates

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M · VALERIVS · M · F
> · LEG · V · ALAVD
ANN · L · STIP · XXVIII
H · S · E

Historical background

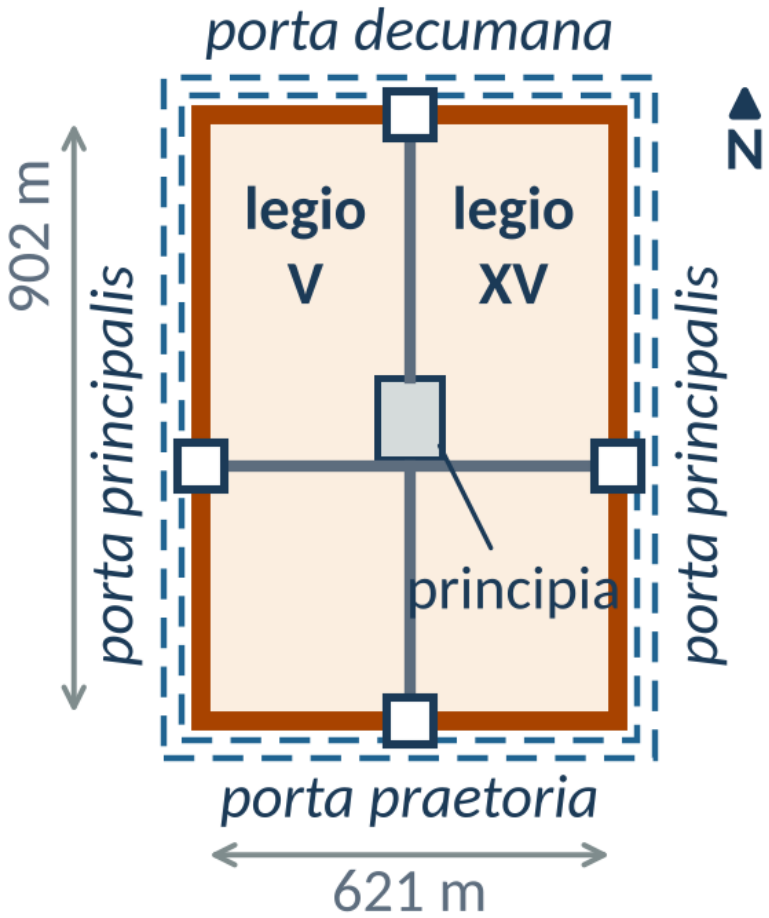
Vetera I · AD 69/70 · Colonia Ulpia Traiana

background for reading at home



Castra Vetera I on the Fürstenberg

- ***castra*** (Latin, plural) = military camp; *castra legionis* = legionary fortress; *castellum* = auxiliary fort; not a "castle"
- **13/12 BC:** Augustan base opposite the Lippe mouth, probably for Drusus' campaigns; earth-and-timber camps
- **AD 14–60:** two legions, rebuilt several times
- **c. AD 60: double-legion fortress** (legio V and XV), 902 m × 621 m ≈ 56 ha, the largest known
- **Lower Germanic Limes:** river frontier (*ripa*), no continuous wall; UNESCO 2021



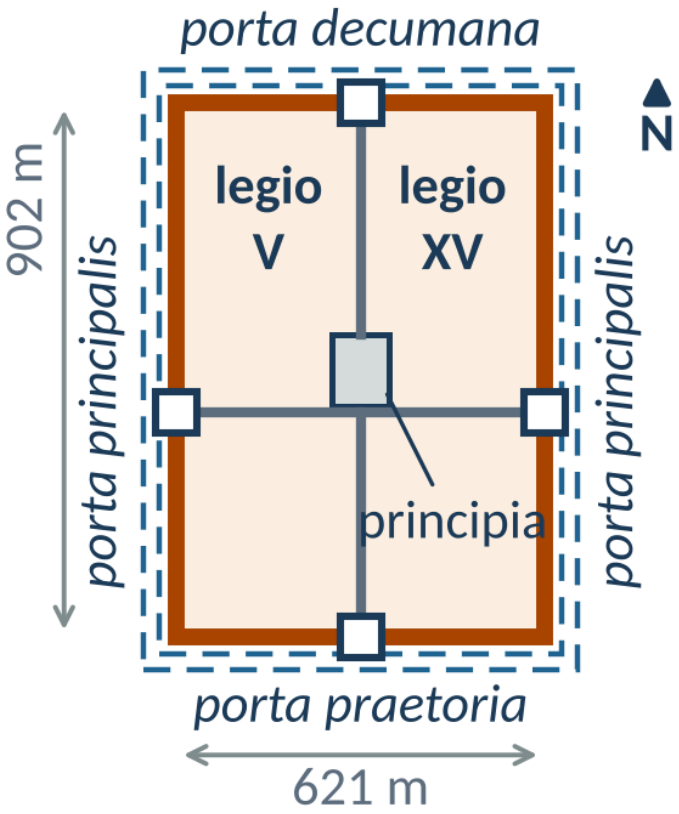
schematic: wall circuit to scale, ditches exaggerated

Wikipedia: Vetera · Lower Germanic Limes · Castra · Vetera (de)

Text: distilled from Wikipedia (CC BY-SA 4.0), see the last slide. Figures: R. Becker (make_slides.py), CC BY-SA 4.0.

Vetera I, c. AD 60: defences and layout

Latin	English	Vetera I, c. AD 60
vallum	rampart, palisade	earlier phases: earth-and-timber
murus	wall, wall walk	3 m, timber frame, clay infill
turres	towers	presumed, not proven
portae	gates	4, each 2 towers + bridge
fossae	V-shaped ditches	2, plus abatis
principia	headquarters	stone, 120 m × 94.8 m
valetudinarium	hospital	stone, 83.5 m × 83.5 m



Wikipedia: Vetera · Castra · Vetera (de)

Text: distilled from Wikipedia (CC BY-SA 4.0), see the last slide. Figures: R. Becker (make_slides.py), CC BY-SA 4.0.

AD 68–70: four emperors, Batavian revolt

Rome: four emperors in one year

	emperor	end
68	Nero †	9 June 68
68–69	Galba	murdered 15 Jan 69
69	Otho	suicide, April 69
69	Vitellius	acclaimed by the Rhine legions; killed 20 Dec 69
69–79	Vespasian	Flavian dynasty

Lower Rhine: revolt of the Batavi

- **Batavi:** Germanic people in the Rhine delta, allies supplying auxiliary troops
- **Julius Civilis:** Batavian noble, Roman citizen, prefect of a cohort
- **69:** revolt during the civil war; Vetera besieged
- **70:** Vetera surrenders (hunger), legions V and XV destroyed, fortress burnt
- **summer 70:** Cerialis defeats Civilis at Vetera

Wikipedia: Year of the Four Emperors · Revolt of the Batavi · Gaius Julius Civilis

Text: distilled from Wikipedia (CC BY-SA 4.0), see the last slide. Figures: R. Becker (make_slides.py), CC BY-SA 4.0. Main ancient source: Tacitus, Histories.

Spring AD 69: at the south gate of Vetera I



- **AI reconstruction:** ChatGPT image, prompts refined with Claude
- **Gate:** after Lehner 1930
- **Tombstone:** fictitious
- **Task:** what is wrong in the picture?

Image: ChatGPT (gpt-image-2); prompts and source checks: Claude (Anthropic). After Lehner 1930, Hanel/Song 2015, Brüggler 2014. Inscription fictitious.

The reconstruction (1): defences and site

Element	In the picture	Evidence
Gate	2 projecting timber towers, 2 passages	excavated post holes: Lehner 1930, Abb. 21–23
Rampart	timber-and-earth wall, timber face	aerial photos: Hanel/Song 2015; Brüggler 2014
Tower	square interval tower	3 × 3 m, c. 67 m apart: Hanel/Song 2015
Ditch	V-shaped, grassed slopes	Lehner 1930, 28–29; Brüggler 2014
Tombstone	stele, portrait niche, early formula	Caelius stone (Xanten); Poblicius (Cologne)
Date	spring AD 69, before the revolt	Tacitus, Histories 4.14, 4.22

Full references in the speaker notes.



The reconstruction (2): soldiers and horse

Element	In the picture	Evidence
Helmet	Imperial-Gallic (Weisenau) type	finds, 1st century: Bishop/Coulston 2006
Armour	segmented plates, belt apron	finds: Bishop/Coulston 2006
Shield	scutum in a leather cover	leather covers from finds (e.g. Valkenburg)
Face	legionary clean-shaven	tombstone portraits, e.g. Caelius (Xanten)
Batavian	mail, spatha, lance, short hair	Bishop/Coulston 2006; Tacitus, Hist. 4.61
Horse	four-horned saddle, no stirrups	Connolly/van Driel-Murray 1991; Capito stone

Assumed, not documented: colours, heights of wall and towers, the persons. Full references in the notes.

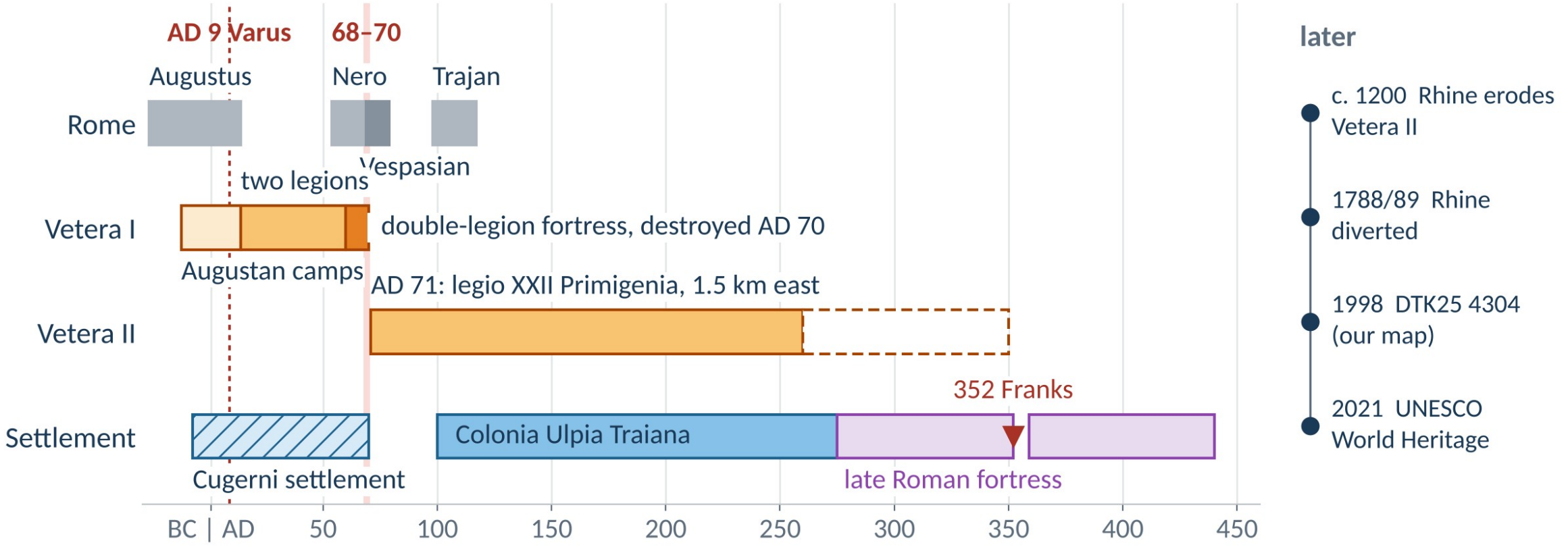
Colonia Ulpia Traiana (Xanten)

- **Before:** settlement of the Cugerni next to Vetera I, probably burnt in AD 70
- **c. AD 100:** Emperor Trajan (Marcus *Ulpius Traianus*) grants the rank of *colonia*, the highest rank of a Roman town
- **73 ha** inside a 3.4 km town wall with 22 towers; c. 10 000 inhabitants; grid of 40 *insulae* with forum, capitol, baths, harbour temple, amphitheatre; harbour on a Rhine arm
- Built partly with stones from the ruins of Vetera I
- **3rd–5th century:** decline; late Roman fortress on the 9 central insulae; taken by the Franks in 352, abandoned in the 5th century
- **Today:** LVR Archaeological Park Xanten; part of the UNESCO World Heritage site since 2021. Find it on the DTK25 photo!

Wikipedia: Colonia Ulpia Traiana · Xanten · Archaeological Park Xanten

Text: distilled from Wikipedia (CC BY-SA 4.0), see the last slide. Figures: R. Becker (make_slides.py), CC BY-SA 4.0.

Xanten in Roman times: a time scale



Wikipedia: Vetera · Colonia Ulpia Traiana · Lower Germanic Limes

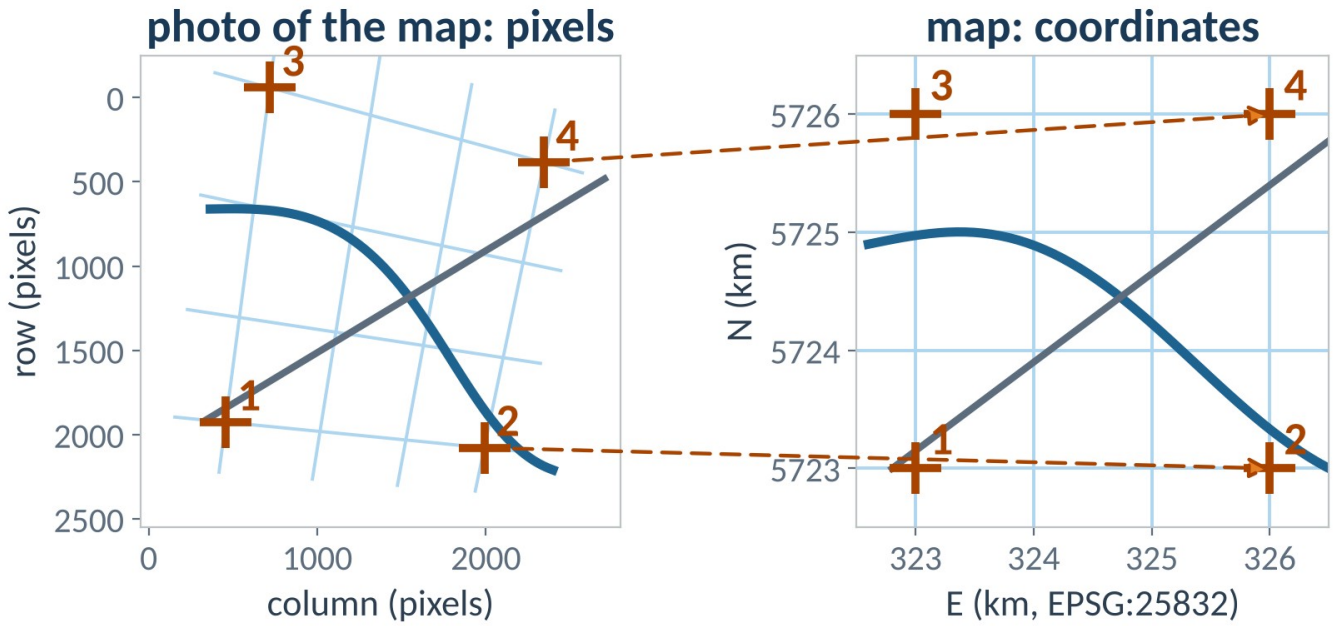
Text: distilled from Wikipedia (CC BY-SA 4.0), see the last slide. Figures: R. Becker (make_slides.py), CC BY-SA 4.0.

The method

control points · transformations · RMS · resampling



What georeferencing does



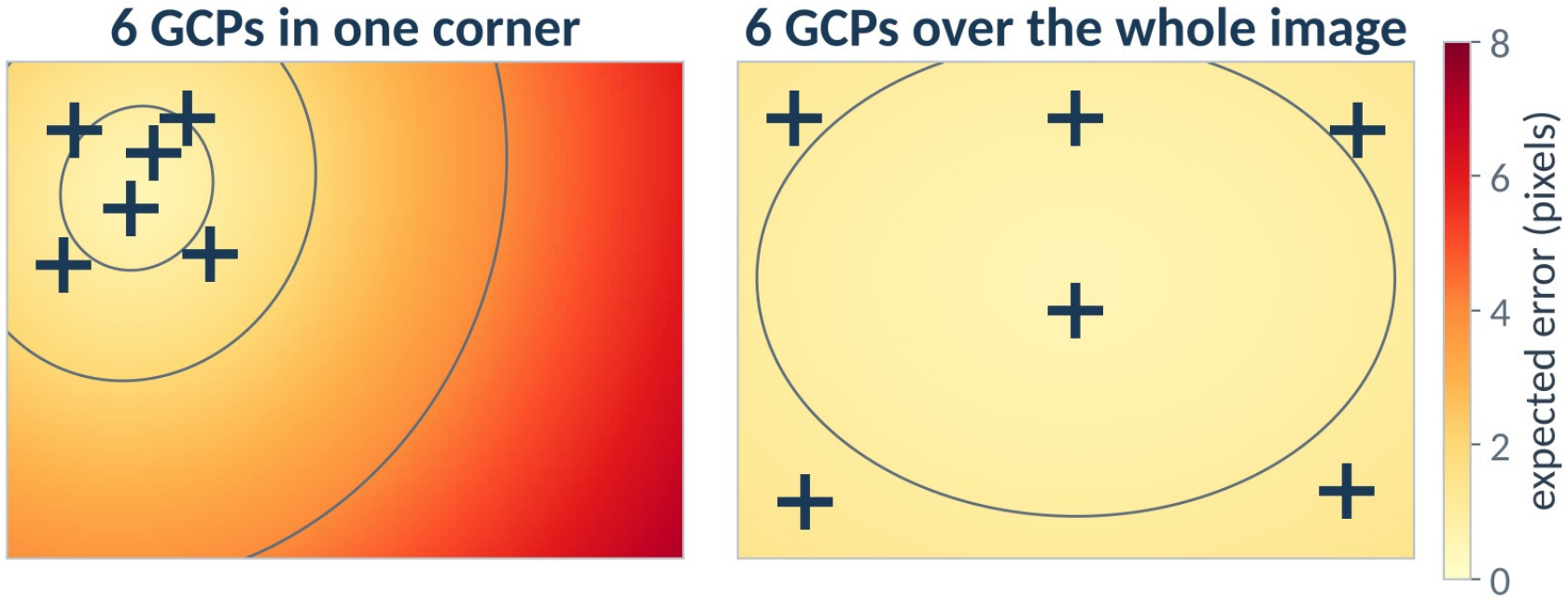
- **Ground control point (GCP):** known in the image (column, row) and on the map (E, N)
- **Transformation:** fitted to the GCPs, applied to every pixel → GeoTIFF in the target CRS

[QGIS user manual: Georeferencer]

Wikipedia: Georeferencing · Ground control point · Image registration · World file · GeoTIFF

Figure: R. Becker (make_slides.py), CC BY-SA 4.0; schematic.

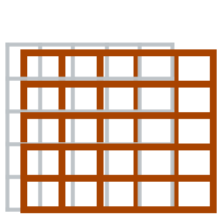
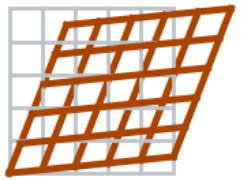
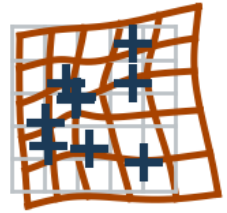
Ground control points: where to click



- **Spread** the GCPs over the whole image, corners included: no extrapolation
- **Sharp and stable:** grid crossings, road crossings, bridges; not rivers or forest edges
- **Enough:** 2–3 × the minimum of the transformation, plus check points

Figures: R. Becker (make_slides.py), CC BY-SA 4.0; simulated example, not the map photo.

Transformations in the Georeferencer

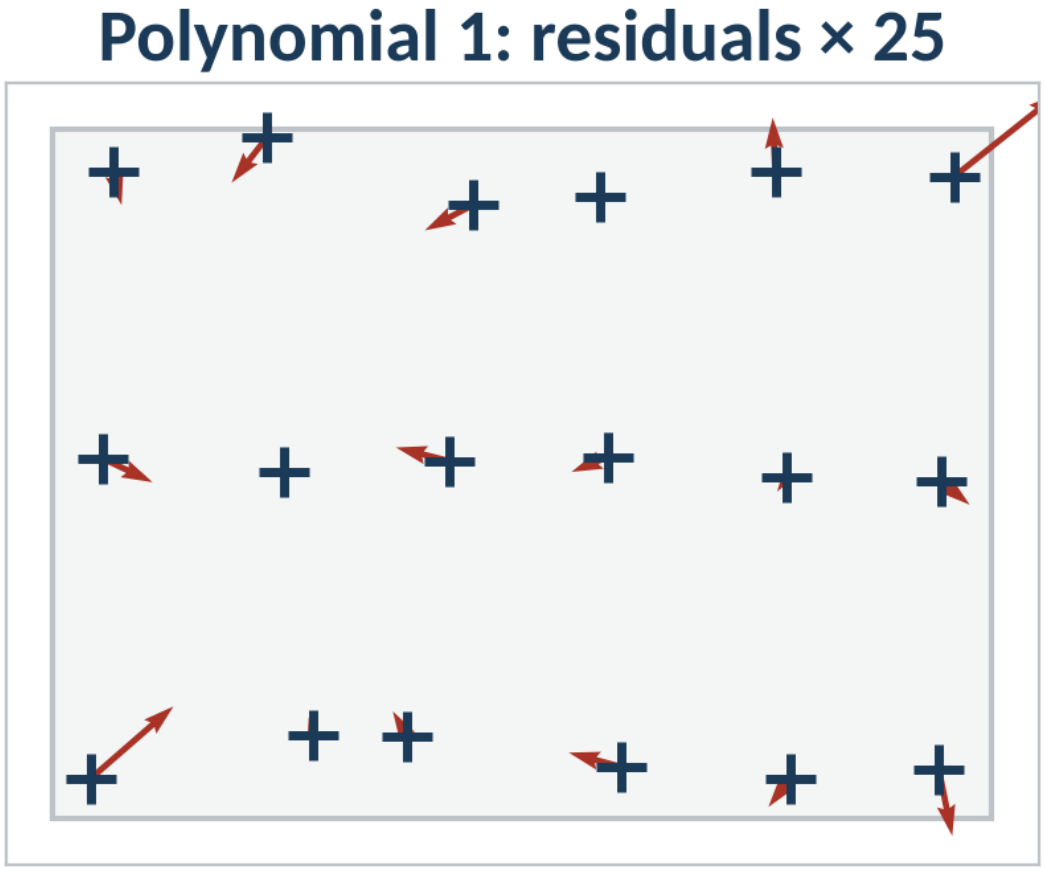
Linear	Helmert	Polynomial 1	Projective	Polynomial 2	Thin Plate Spline
					
shift + scale ≥ 2 GCPs	+ rotation ≥ 2 GCPs	+ shear (affine) ≥ 3 GCPs	tilted photo ≥ 4 GCPs	bends lines ≥ 6 GCPs	rubber sheet ≥ 10 GCPs

- **More parameters** fit the GCPs better, but need more of them and can bend the image
- **Choose by the source:** scan of a good map → Helmert / Polynomial 1; photo taken at an angle → Projective; warped paper → Polynomial 2; poor originals → Thin Plate Spline

Wikipedia: Helmert transformation · Affine transformation · Homography · Thin plate spline

Text: QGIS user manual, Georeferencer (docs.qgis.org, CC BY-SA 3.0). Figures: R. Becker (make_slides.py), CC BY-SA 4.0.

Residuals and RMS error



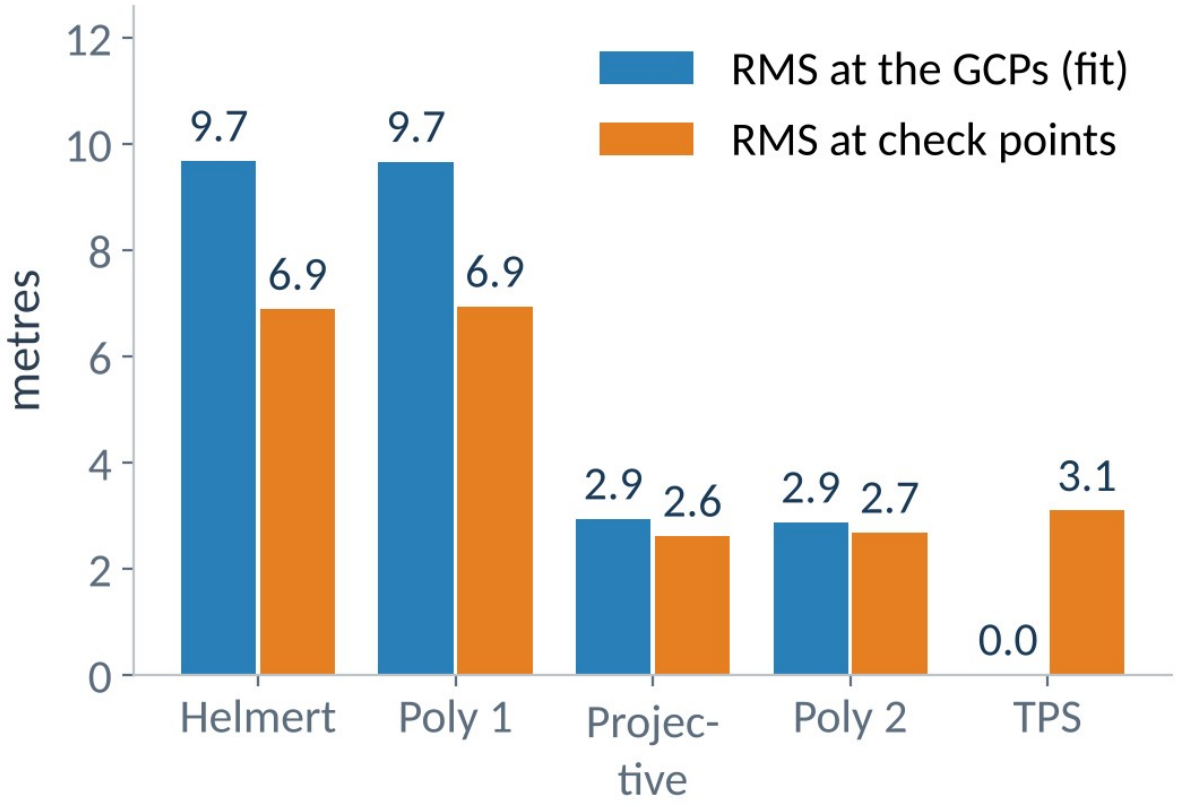
$$RMS = \sqrt{[\ \Sigma\ (dX^2 + dY^2)\ /\ n\ ]}$$

- **Residual:** how far a GCP lands from its map position
- **RMS:** one number for all GCPs, in pixels or metres
- **Systematic pattern** → transformation too simple
- **One long arrow** → wrong click: fix it or switch the GCP off

Wikipedia: Errors and residuals · Root-mean-square deviation · Least squares

Figures: R. Becker (make_slides.py), CC BY-SA 4.0; simulated example, not the map photo.

A small RMS is not yet an accurate map

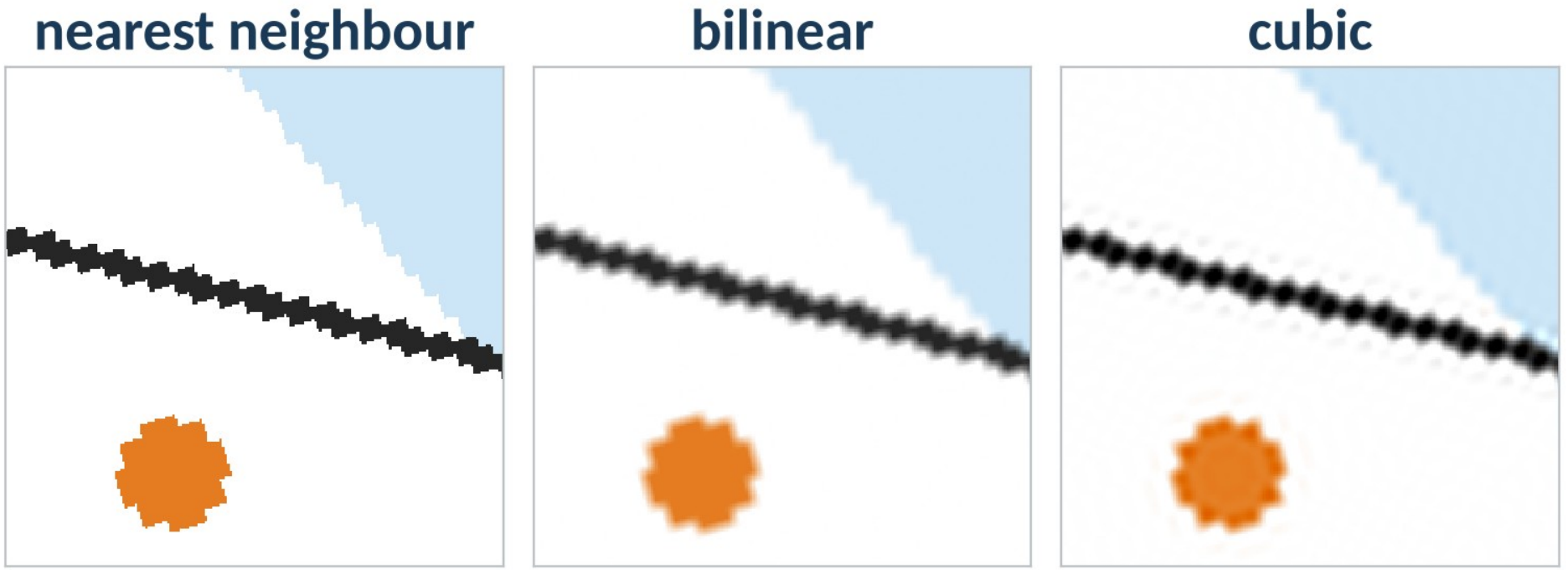


- **RMS at the GCPs** shrinks with every extra parameter
- **Thin Plate Spline (TPS):** RMS = 0 by construction
- **Check points:** switch 3–5 GCPs off and look at their error
- **Report:** transformation, number of GCPs, RMS and check error in metres

Wikipedia: Overfitting · Cross-validation

Figures: R. Becker (make_slides.py), CC BY-SA 4.0; simulated example, not the map photo.

Resampling: new pixels from old ones



- **Nearest neighbour:** keeps the original values; required for classes and codes
- **Bilinear, cubic:** smoother, but mixed values; fine for photos and map scans

Wikipedia: Nearest-neighbor interpolation · Bilinear interpolation · Bicubic interpolation

Figure: R. Becker (make_slides.py), CC BY-SA 4.0; a synthetic map patch rotated by 17°, ×5.

Optional · code

Georeferencing step by step

```
import numpy as np, pandas as pd
g = pd.read_csv("dtk_photo.points", comment="#")   # QGIS GCP file
A = np.c_[g.sourceX, g.sourceY, np.ones(len(g))]    # Polynomial 1
xy = g[["mapX", "mapY"]].to_numpy()
coef, *_ = np.linalg.lstsq(A, xy, rcond=None)
print("RMS", np.sqrt(((A @ coef - xy) ** 2).sum(1).mean()))
```

```
gdal_translate -gcp 103.1 402.4 322000 5728000 ... a.jpg a_gcp.tif
gdalwarp -t_srs EPSG:25832 -order 1 -r cubic a_gcp.tif a_utm.tif
# -order 2: Polynomial 2,  -tps: Thin Plate Spline
```

Same mathematics as the Georeferencer: least squares on the GCPs, then warp and resample (as in geo0162). Extend: leave-one-out check error.

Code: R. Becker (CC BY-SA 4.0); GDAL utilities gdal_translate, gdalwarp.



Take-aways: georeferencing

- **Know the CRS of your GCPs:** read the map legend (grid, datum)
- **Spread sharp, stable GCPs** over the whole image
- **Choose the transformation by the source,** not by the smallest RMS
- **Judge with check points** and by overlaying a reference map
- **Resample with nearest neighbour** when the values are classes; cubic for photos
- **Keep the .points file:** it documents and repeats your work



Check yourself

1 Your Thin Plate Spline shows RMS = 0 m. Is your map perfect?

2 One GCP has a residual of 150 m, all others less than 30 m. What do you do?

3 You georeference a land-cover map with class codes. Which resampling?

4 Why are the UTM grid crossings better GCPs than road crossings on our map?



Check yourself: answers

1 No. TPS passes through every GCP by construction; only check points tell you the accuracy.

2 Check that point: wrong crossing clicked, wrong margin number or a typo. Fix it or untick it.

3 Nearest neighbour: it keeps the original codes; bilinear or cubic would invent new values.

4 Their coordinates are printed in the margin (no reference map needed), they are sharp and they cover the whole image evenly.



Case study: Castra Vetera

TK25 4304 Xanten (1998) · Vetera I reconstruction · DTM



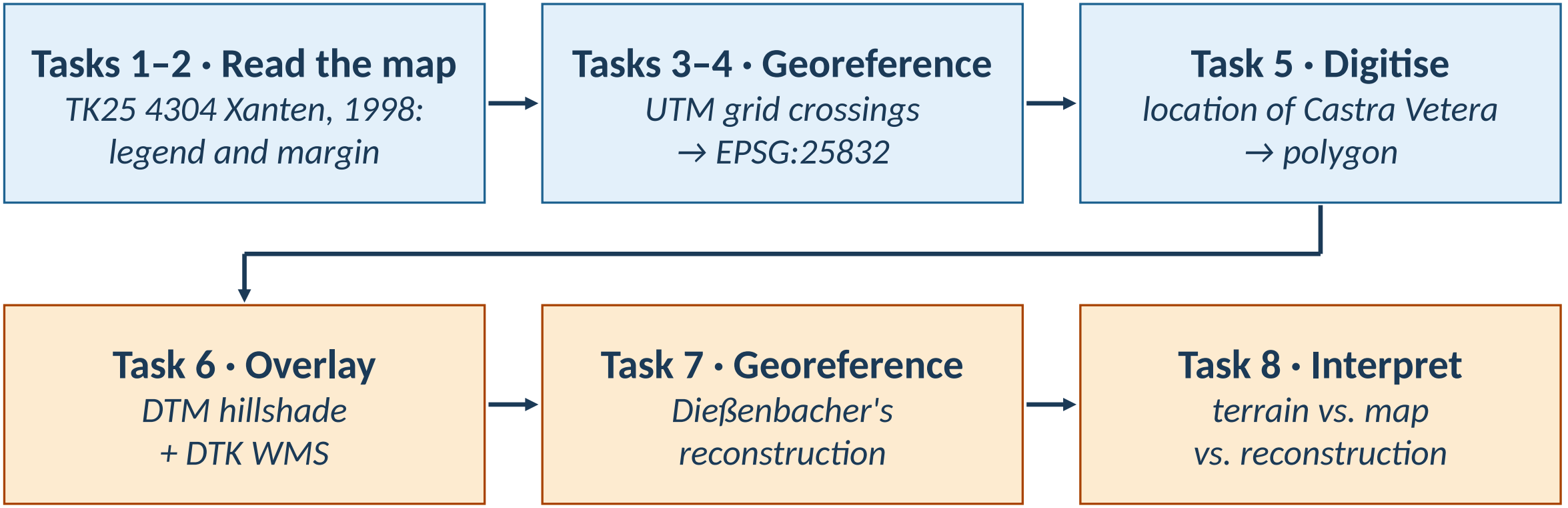
Your path through the case study

Task	What you do	Time
1–2	Read the map: legend and margin	30 min
3	Georeference the photo with the UTM grid crossings	45 min
4	Compare transformations, use check points	30 min
5	Digitise the location of Castra Vetera	20 min
6	Read the DTM hillshade	20 min
7–8	Georeference and interpret the reconstruction	50 min

Your files: in the session folder where you found this deck: the task sheet, the QGIS project and, in its data/ folder, all images (laptop: git pull first).

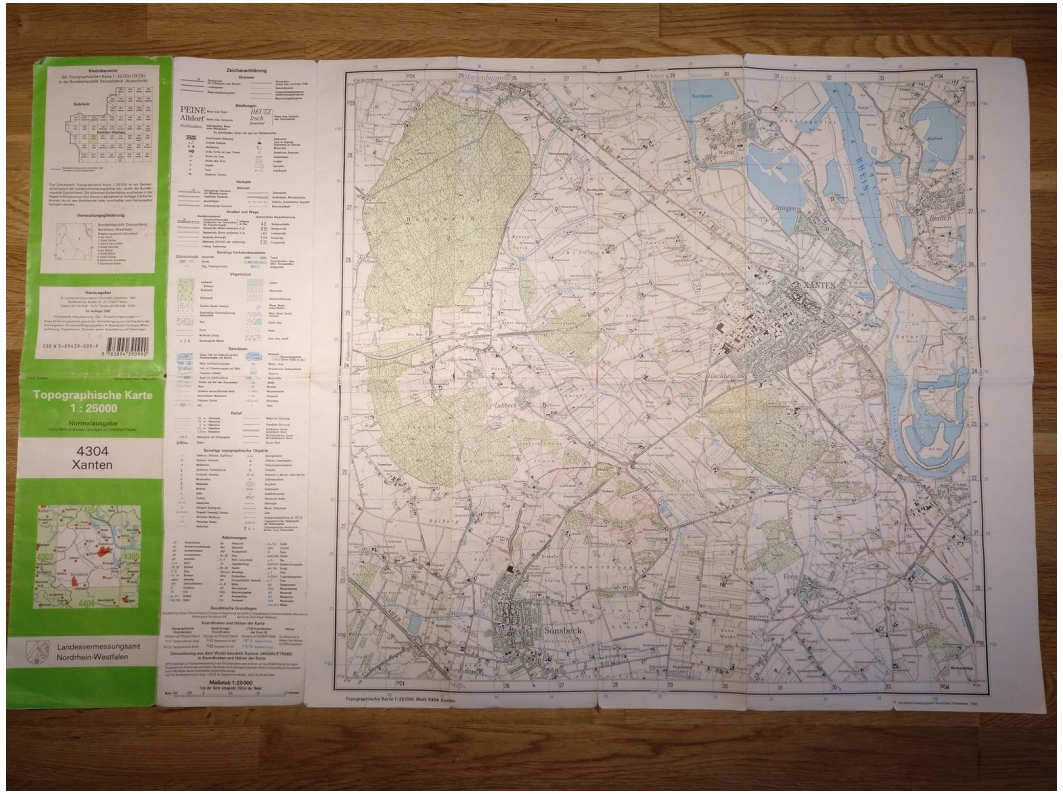
Not finished in class? Continue at home with this deck and the task sheet geo0241_Georeferencing_Castra_Vetera.md. The task number is shown top right on every slide.

The processing chain



Tool: QGIS Georeferencer (Layer → Georeferencer…), digitising in a GeoPackage

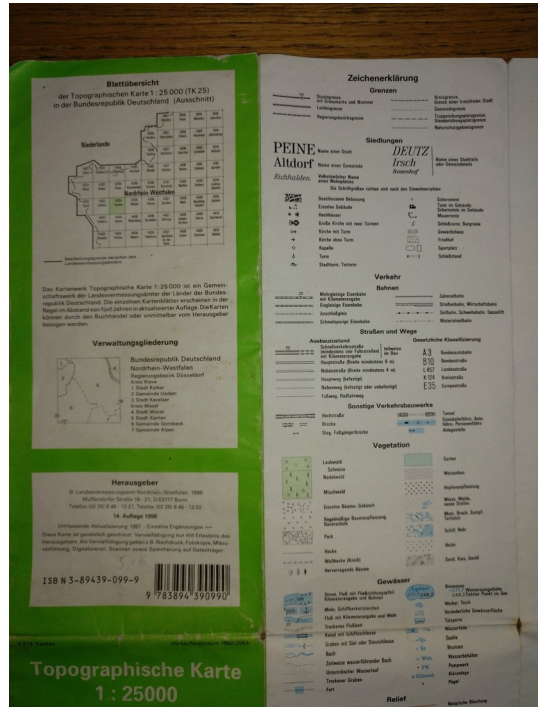
Material: the paper map TK25 4304 Xanten



whole sheet, cover and legend



geodetic basis



publisher, edition

Files: data/geo0241_DTK25_4304_Xanten_1998_*.jpg (Sheet, Geodetic_Basis, Publisher)

Photos: R. Becker (2018) of TK25 4304 Xanten, 14. Auflage 1998, © Geobasis NRW (formerly Landesvermessungsamt NRW), dl-de/zero-2-0.

Investigate the map

- **Type of map:** which map series, scale, edition name?
- **Publisher:** who made it? Who is responsible today?
- **Year:** edition, last update, sales period – how current is the content?
- **CRS:** geodetic datum, ellipsoid, projection of the map content?
- **Grids:** which coordinate grids are printed, and in which CRS? Which EPSG code do you use in the Georeferencer?
- **Look at:** map margin, cover and the legend panels Geodätische Grundlagen and Koordinaten
- **Where:** the photos in data/ of your session folder – open them at full size and zoom in; the slides only show previews

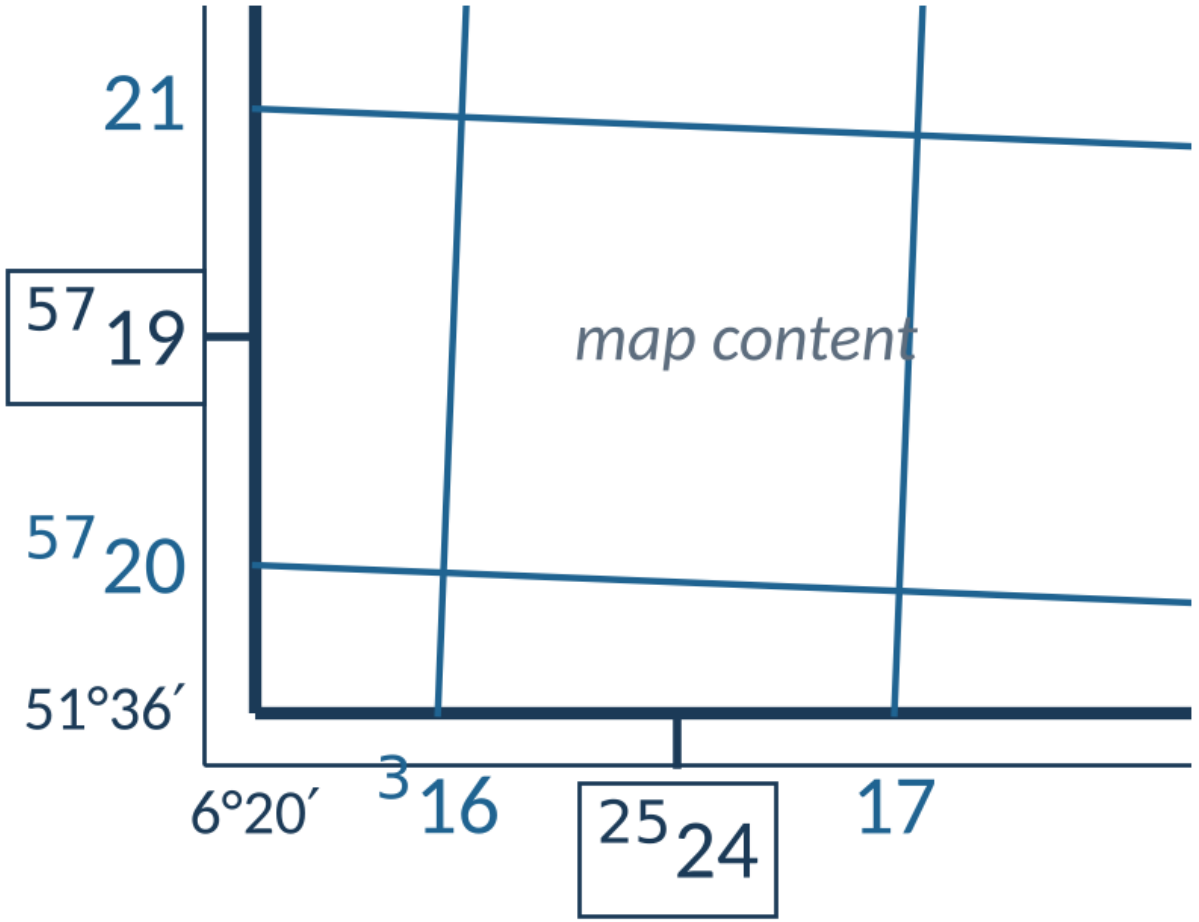


Material: the south-west corner of the sheet



Photos: R. Becker (2018) of TK25 4304 Xanten, 14. Auflage 1998, © Geobasis NRW (formerly Landesvermessungsamt NRW), dl-de/zero-2-0.

Read the map margin



- Which lines belong to the UTM grid?
- What do the numbers mean? What are the small raised digits?
- What is the interval between two lines – in km and in cm on the map?
- And the boxed numbers and the corner values?

Wikipedia: Universal Transverse Mercator · Gauss-Krüger (de)

Figure: R. Becker, schematic after the map margin (not to scale), CC BY-SA 4.0.

Our map has three coordinate systems

on the map sheet	datum	EPSG	for the GCPs
UTM zone 32, km grid	ETRS89 (“WGS84/ETRS89”)	25832	yes: grid crossings
Gauss-Krüger, zone 2	DHDN (Potsdam, Bessel)	31466	only with 31466
geographic, ° ′ ″	DHDN (Potsdam, Bessel)	4314	never as 4326

- **Read the legend first:** which grid, which datum? The sheet itself says how to convert GNSS positions: +2.5″ longitude, +4.8″ latitude
- **Same numbers, wrong datum:** about 155 m off (geo0220, DHDN → ETRS89)
- **Our choice:** UTM grid crossings at full kilometres, target CRS EPSG:25832

Wikipedia: Universal Transverse Mercator · Gauß-Krüger-Koordinatensystem (de) · Deutsches Hauptdreiecksnetz (de)

Source: legend of DTK25 4304 Xanten, Landesvermessungsamt NRW, 1998 (facts only).

Material: detail of the Fürstenberg

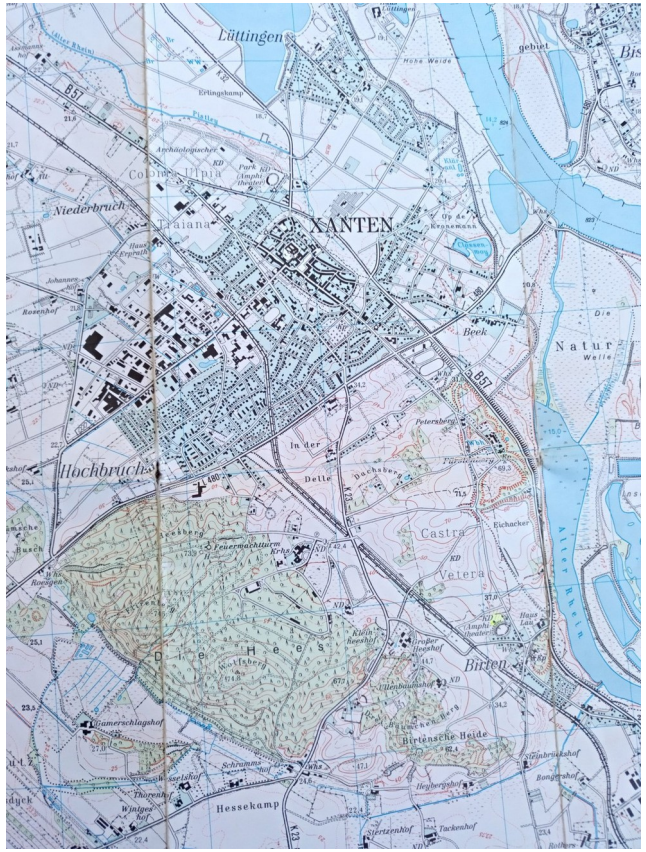
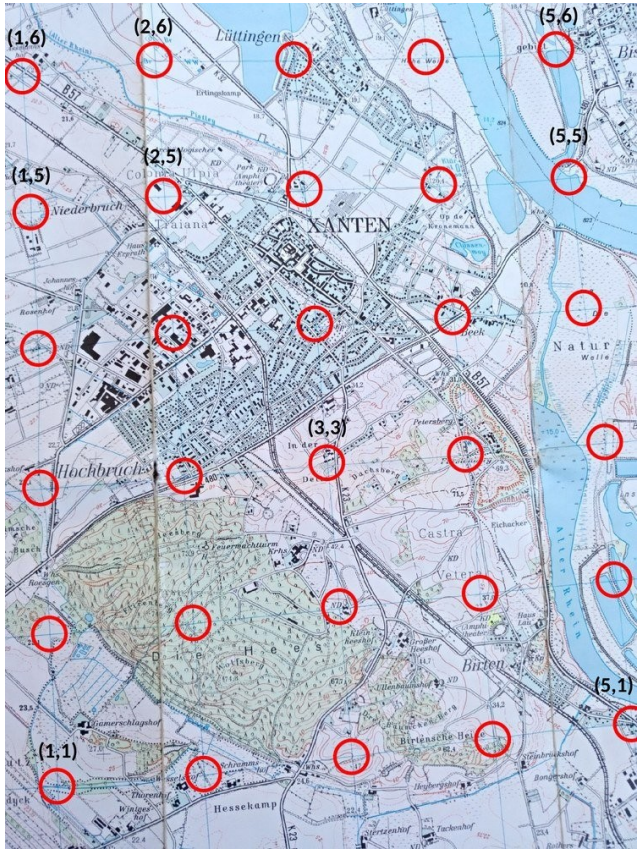


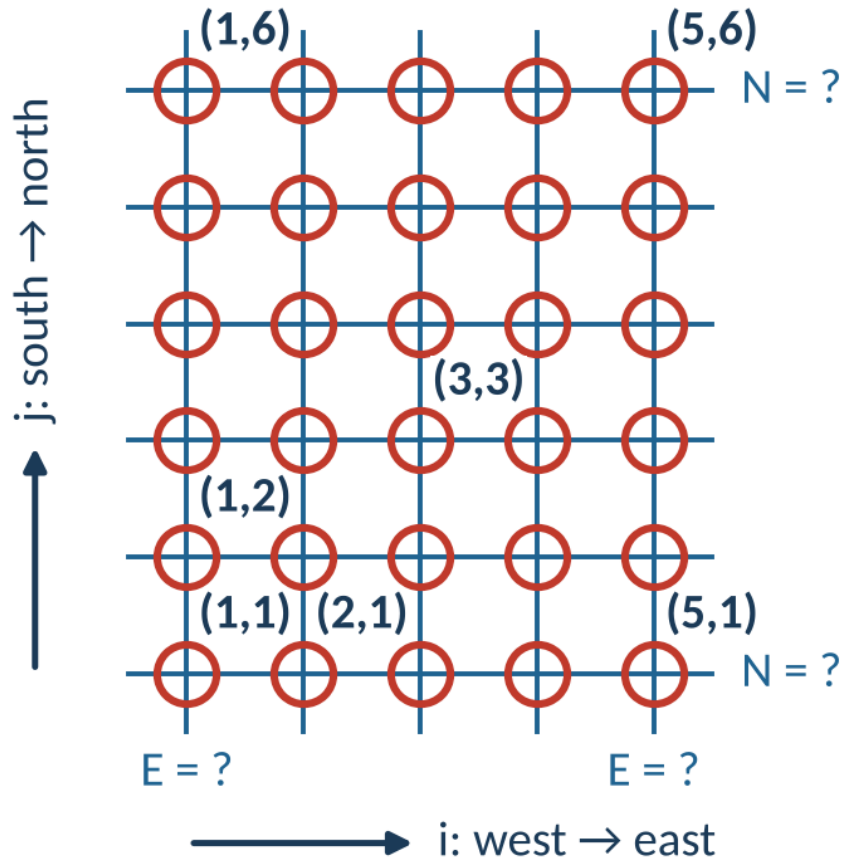
photo to georeference



grid crossings (i, j), (1,1) = lower left

Photos: R. Becker (2018) of TK25 4304 Xanten, 14. Auflage 1998, © Geobasis NRW (formerly Landesvermessungsamt NRW), dl-de/zero-2-0.

Ground control points: the grid crossings



- The red circles number the crossings (i, j): i = column from the west, j = row from the south; (1,1) is the lower left crossing (smallest E and N)
- Determine (E, N) of every crossing from the blue numbers in the margin
- Enter them in the Georeferencer, target CRS EPSG:25832
- Compare transformations and their RMS error

Figure: R. Becker, CC BY-SA 4.0.

Worked example: crossing (1,1)

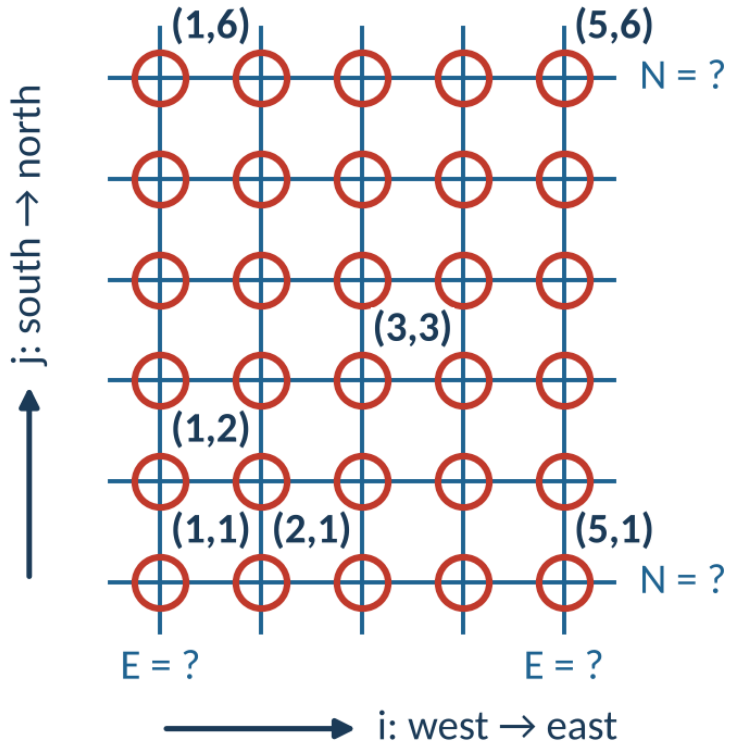
1 Find it: (1,1) is the lower left red circle of the Notations file, near Thorenhof

2 Easting: follow its blue grid line down to the bottom margin of the sheet photo: blue “22”. With the raised “3” of the first line ($^{3}16$): E = 322 km = 322 000 m

3 Northing: follow the blue line to the side margin: blue “23”, with the raised “57” of $^{57}20$:
N = 5 723 km = 5 723 000 m

4 Next crossings: one blue line further = 1 km = 1000 m: east → E + 1000, north → N + 1000

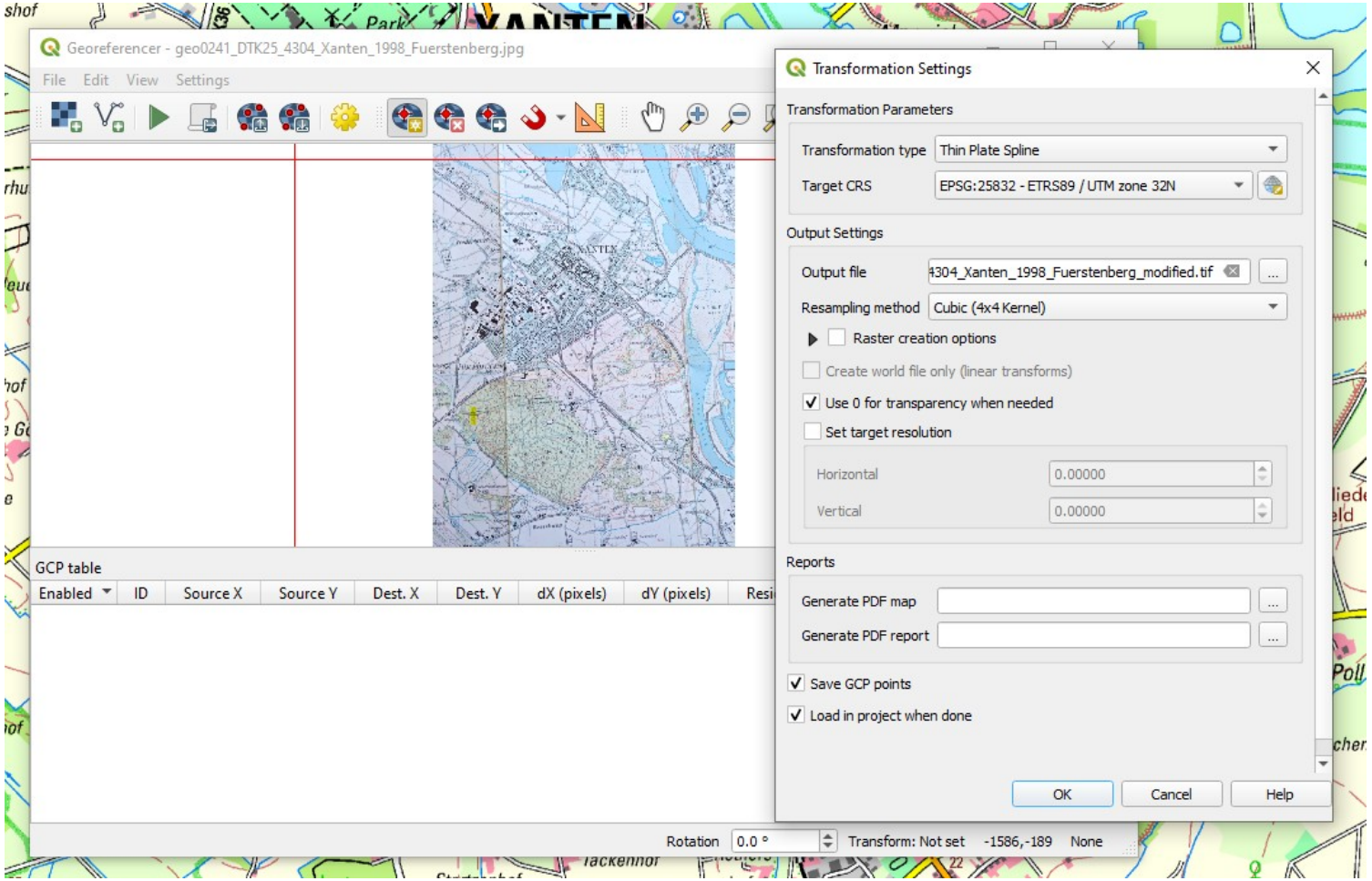
Careful: black numbers are Gauss-Krüger, not UTM



(1,1): E 322 000, N 5 723 000

Figure: R. Becker, CC BY-SA 4.0. Margin numbers: TK25 4304 Xanten (1998), © Geobasis NRW, dl-de/zero-2-0.

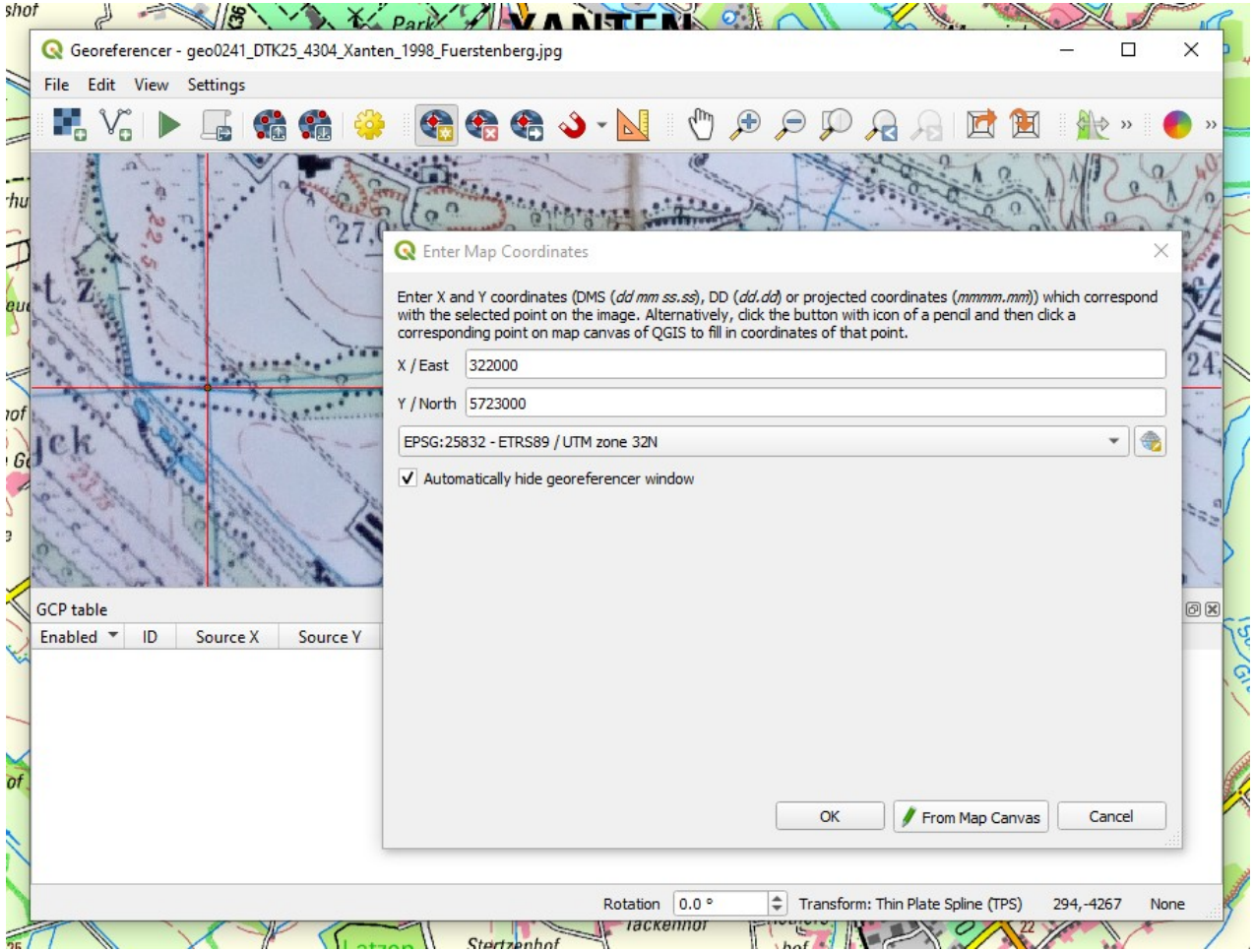
In QGIS (1): open the photo, transformation settings



- Layer → Georeferencer… → Open Raster: the photo
- Settings: type (compare several in task 4), EPSG:25832, output GeoTIFF, Cubic
- Tick “Save GCP points” and “Load in project when done”

Screenshot: QGIS 3.44, R. Becker. Background © OpenStreetMap contributors (ODbL); map photo © Geobasis NRW (dl-de/zero-2-0).

In QGIS (2): add a ground control point



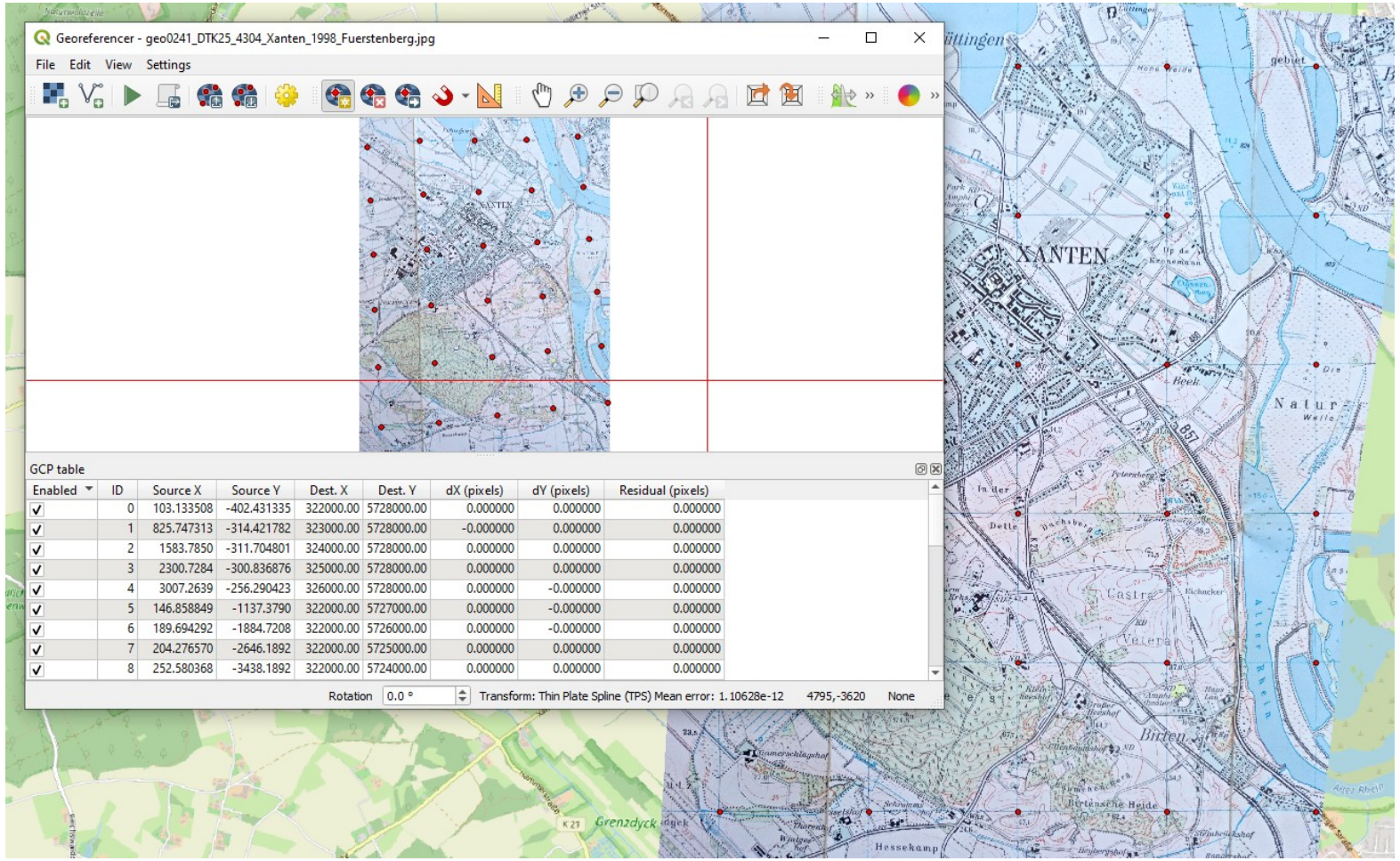
Add Point, click the point in the photo (zoom in). Then:

1 Known? Type E and N in metres: our grid crossings

2 Not known? “From Map Canvas”: click the same feature on the map (task 7)

Screenshot: QGIS 3.44, R. Becker. Background © OpenStreetMap contributors (ODbL); map photo © Geobasis NRW (dl-de/zero-2-0).

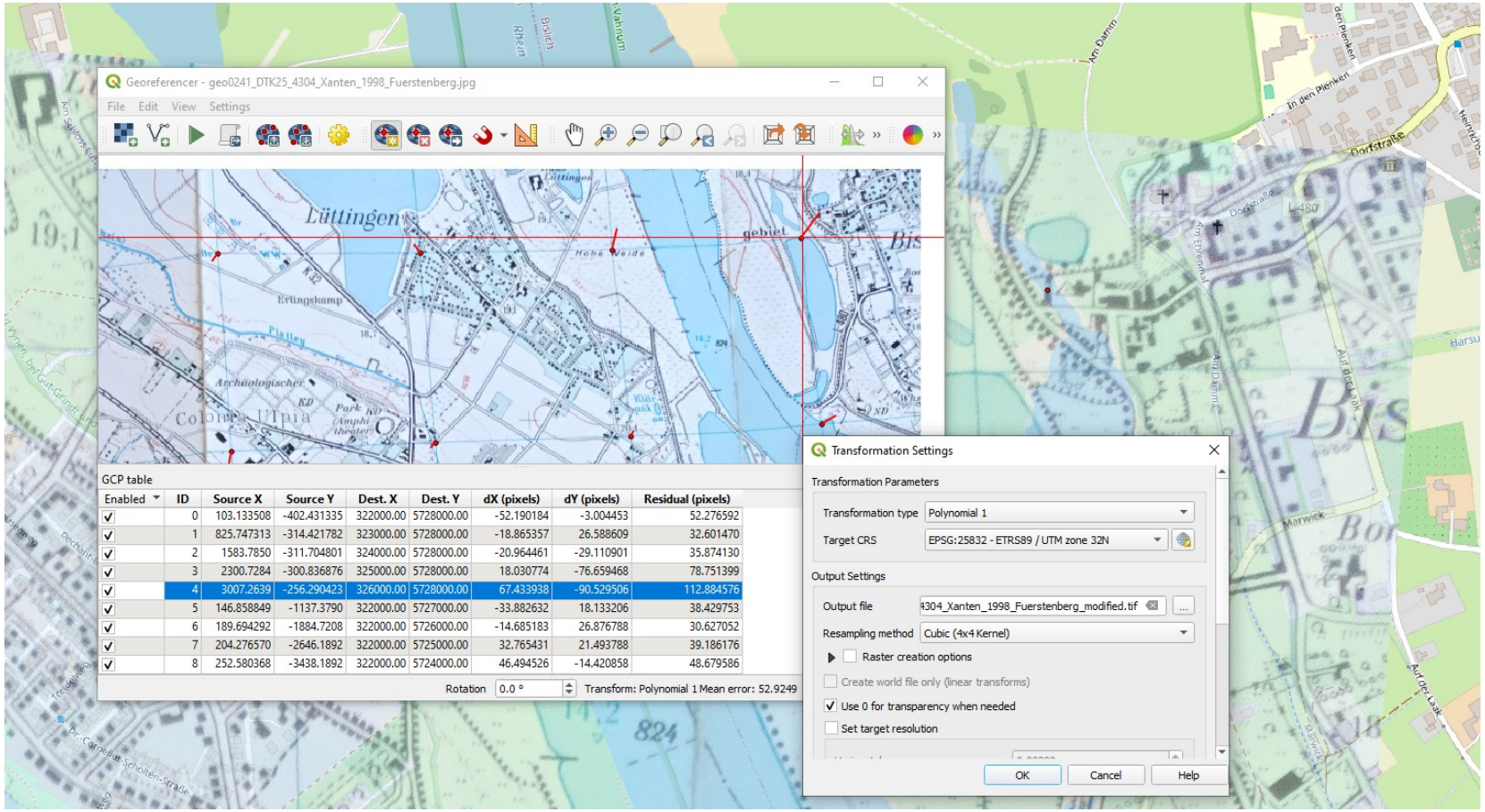
In QGIS (3): GCP table and the result on the map



- GCP table: dX, dY, residual per point; mean error below
- Thin Plate Spline: all residuals 0 – tells you nothing. Untick 3–5 as check points
- Behind: the result on the map. Do roads and the Rhine fit?

Screenshot: QGIS 3.44, R. Becker. Background © OpenStreetMap contributors (ODbL); map photo © Geobasis NRW (dl-de/zero-2-0).

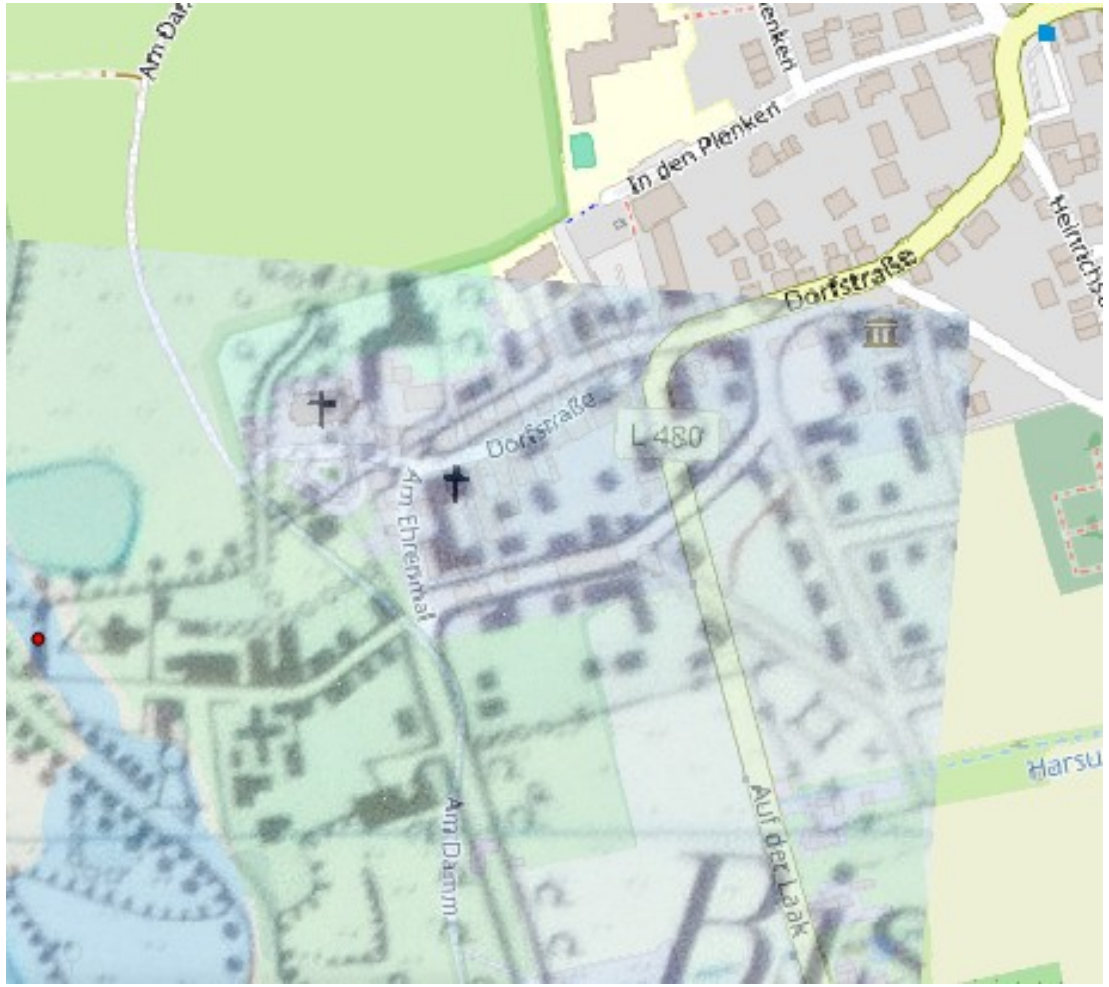
In QGIS (4): Polynomial 1 – residuals



- Same 30 crossings: mean error 52.9 px ≈ 70 m
- Upper right crossing (5,6), highlighted: dX = +67 px, dY = −91 px
- Rows count downwards: the red arrow points up and right, 113 px ≈ 155 m

Screenshot: QGIS 3.44, R. Becker. Background © OpenStreetMap contributors (ODbL); map photo © Geobasis NRW (dl-de/zero-2-0).

In QGIS (5): Polynomial 1 – the shift on the map



- TK25 (semi-transparent) over OpenStreetMap, near the upper right crossing
- Find the bend from Auf der Laak (south) to Dorfstraße (east): yellow in OSM, dark in the TK25
- The TK25 lies about 150 m south-west – the arrow pointed north-east: same misfit, seen from the photo and from the map
- Why is the misfit largest at the corner? Explain it in your report

Screenshot: QGIS 3.44, R. Becker. Background © OpenStreetMap contributors (ODbL); map photo © Geobasis NRW (dl-de/zero-2-0). Detail of the QGIS map canvas.

Am I on track? Expected values

Transformation	RMS with all 30 crossings
Helmert, Polynomial 1	about 50–80 m
Projective, Polynomial 2	about 15–30 m
Thin Plate Spline	0 m – tells you nothing

Rule of thumb: a residual above 3 × RMS points to a wrong click or a wrong coordinate

- Your values differ a little: your clicks are not ours
- Much larger? Look for one GCP with a far larger residual than the others
- Why are some transformations so much better here? Explain it in your report

Values: reference run by R. Becker with the 30 grid crossings (2026), rounded.



Troubleshooting

What you see	Likely cause	What to do
Result in Africa or the sea	target CRS not set (EPSG:4326 or project CRS)	Transformation Settings → EPSG:25832
Result 2–5 km off	black Gauss-Krüger numbers read instead of blue UTM ones	use the blue numbers, see the worked example
Image mirrored or turned	E and N swapped for some points	E ≈ 32x xxx, N ≈ 5 72x xxx
One huge residual	wrong crossing clicked or a typo	check (i, j) and the numbers, or untick it
RMS = 0	Thin Plate Spline: fits every point	judge with check points
No output, or not found	output file or folder not set	set the output GeoTIFF in your session folder



Castra Vetera on the map: 1998 and today



TK25 4304, 1998 (photo)



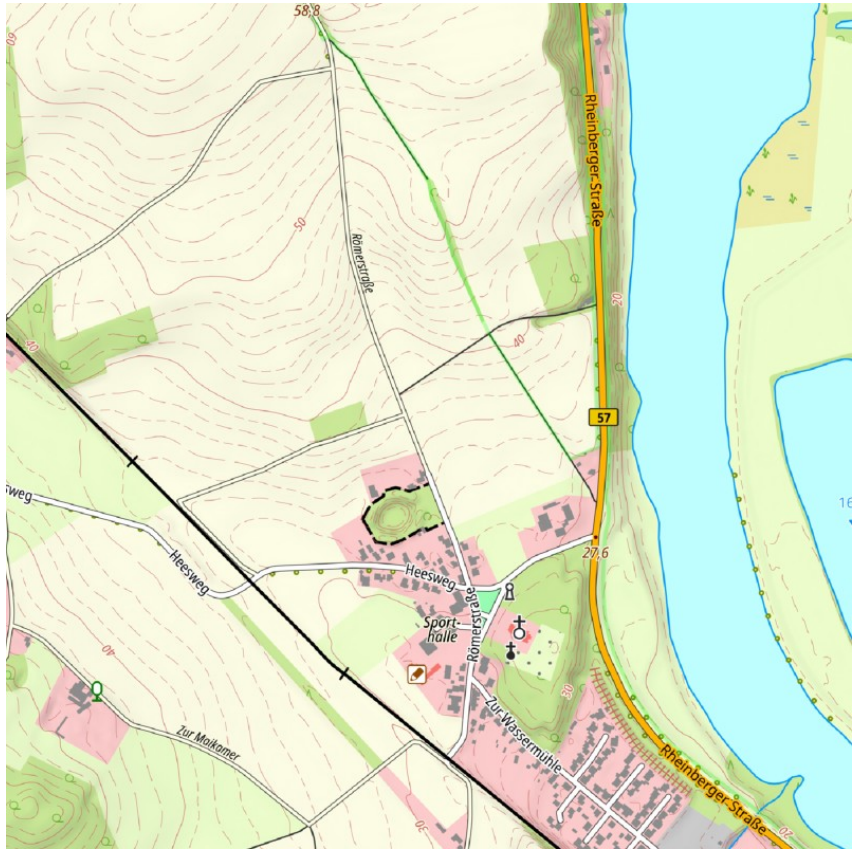
DTK25 today, 1:25 000 (© Geobasis NRW)

Photos: R. Becker (2018) of TK25 4304 Xanten, 14. Auflage 1998, © Geobasis NRW (formerly Landesvermessungsamt NRW), dl-de/zero-2-0.

Today's DTK: where is Castra Vetera?



1:25 000



about 1:6 000 (zoomed in)

- 1998: label and KD mark the approximate location
- What is left today?
- Digitise the location from the 1998 map

© Geobasis NRW (dl-de/zero-2-0), WMS. Left 2.8 km × 2.8 km, right 1.4 km × 1.4 km, EPSG:25832.

Task: read the DTM hillshade



- Overlay your georeferenced map and the polygon of Castra Vetera with the DTM hillshade
- Which pronounced features does the hillshade show around the polygon? Describe them
- Which of them do you find on the 1998 map, which on today's DTK?

Wikipedia: Hillshading · Digital elevation model

DTM hillshade: © Geobasis NRW (dl-de/zero-2-0), WMS (wms_nw_dgm-schummerung), 2.8 km × 2.8 km.

Material: Dießenbacher's reconstruction of Vetera I



- No grid: find features you also see on your georeferenced map or in the hillshade
- Set them with “From Map Canvas” (option 2)
- Few GCPs: choose a simple transformation

Reconstruction of Vetera I: Dr. Frank Dießenbacher, Dießenbacher Informationsmedien, Xanten (www.diessenbacher.com), CC BY 4.0.

Task: interpret the reconstruction

- Georeference the reconstruction: GCPs from your georeferenced map and the DTM
- Overlay it with the DTM hillshade and the DTK
- How does the reconstruction imagine the surroundings of Castra Vetera?
- Find inconsistencies between the reconstruction and the terrain. What could be the reasons?

Hand in: the report described in the task sheet geo0241_Georeferencing_Castra_Vetera.md



Sources and further reading

Castra Vetera I on the Fürstenberg: Vetera · Lower Germanic Limes · Castra · Vetera (de)

Vetera I, c. AD 60: defences and layout: Vetera · Castra · Vetera (de)

AD 68–70: four emperors, Batavian revolt: Year of the Four Emperors · Revolt of the Batavi · Gaius Julius Civilis

Colonia Ulpia Traiana (Xanten): Colonia Ulpia Traiana · Xanten · Archaeological Park Xanten

Xanten in Roman times: a time scale: Vetera · Colonia Ulpia Traiana · Lower Germanic Limes

What georeferencing does: Georeferencing · Ground control point · Image registration · World file · GeoTIFF

Transformations in the Georeferencer: Helmert transformation · Affine transformation · Homography · Thin plate spline

Residuals and RMS error: Errors and residuals · Root-mean-square deviation · Least squares

A small RMS is not yet an accurate map: Overfitting · Cross-validation

Resampling: new pixels from old ones: Nearest-neighbor interpolation · Bilinear interpolation · Bicubic interpolation

Read the map margin: Universal Transverse Mercator · Gauss-Krüger (de)

Our map has three coordinate systems: Universal Transverse Mercator · Gauß-Krüger-Koordinatensystem (de) · Deutsches Hauptdreiecksnetz (de)

Task: read the DTM hillshade: Hillshading · Digital elevation model

Text distilled from Wikipedia (CC BY-SA 4.0) and the QGIS user manual (CC BY-SA 3.0), October 2026. Figures: R. Becker (make_slides.py), CC BY-SA 4.0.



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